

Phased Array Systems – Design Considerations & System Demonstration

Connor Pope
Analog Devices
UMass Lowell
Boston, MA

connor.pope@analog.com

Hong Tang
Electrical and Computer
Engineering Dept.
University of Massachusetts
Lowell
Lowell, MA, USA

Bowen Zheng
Electrical and Computer
Engineering Dept.
University of Massachusetts
Lowell
Lowell, MA, USA

Hualiang Zhang
Electrical and Computer
Engineering Dept.
University of Massachusetts
Lowell
Lowell, MA, USA
hualiang_zhang@uml.edu

Abstract—This paper discusses the design, verification, and calibration of phased array platforms at a high level. The phased array architectures (analog, digital, hybrid) and their trade-offs are discussed. Antenna selection and design options are considered based on application requirements. Measurement-based cascade models of the mixed-signal channels are described to obtain system figures of merit as a function of scan angles and amplitude tapers. Limitations imposed by quantization and root-mean square errors can be modelled using these cascades. Beam weight synthesis and array thinning are shown to achieve desired beam masks, contours, and power consumption specifications for arbitrary arrays with measured, modeled, or ideal embedded element patterns. Finally, a basic calibration routine for a hybrid phased array architecture is described, and the simulated and measured performance is compared. The application requirements of a phased array system drive antenna selection, aperture size, lattice shape, power consumption, calibration capability, and the number of simultaneous beams available.

Keywords—Wideband Phased Array, Architecture, Antenna, Simulation, Test, Measurement, Calibration

I. INTRODUCTION

Phased array systems can come in various architectures, each implementation is potentially unique, and can be highly complex. Fundamentally, a phased array system is made up of an array of antenna elements, a mixed-signal channel which transmits and/or receives at each element, and a computer to process the information and set the system state for scanning and amplitude tapering. An analog phased array architecture requires phase control in the analog channel of each individual antenna element to accomplish analog beamforming. A digital phased array architecture applies phase control via post-processing through digital beamforming (DBF), allowing for any number of digital beams. How each of these are implemented depends on a large trade-space driven by application requirements.

Phased arrays increase the efficiency of antenna platforms by increasing radiation in a target direction, and drastically reducing radiation in all other directions. This increases signal-to-noise ratio (SNR) at the target for the same input power and physical channel, while reducing interference from nearby platforms or interferers.

A current motivation to study phased arrays, particularly digital and hybrid, is in their potential for multi-application use and the implementation of adaptive beamforming techniques. Current phased array technologies are restricted by their architecture to a single-application or are not easily reconfigurable to other applications. Figure 1 shows an example of size, weight, power, and cost (SWaP-C) being reduced because of a single multi-function phased array replacing multiple single-function phased arrays.

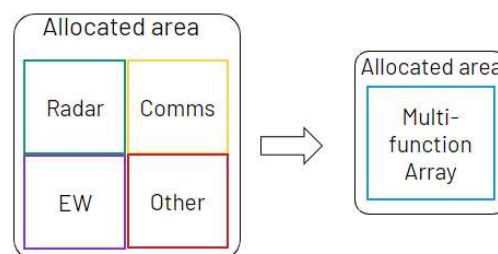


Figure 1. SWaP-C improvement when going from single-function phased arrays to one multi-function phased array.

Different applications have different requirements for the phased array design. Common phased array applications and their requirements are shown below in Figure 2.

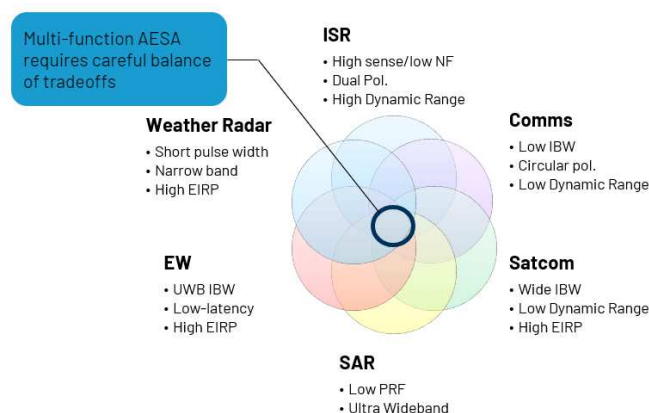


Figure 2. Application space requirements for phased arrays.

First, trade-offs of phased array architectures are described. Second, antenna selection and design are discussed with a focus

on the embedded element pattern and mutual coupling parameters. Third, the mixed-signal channel and its measurement-based model for system-level verification and calibration limitations are discussed. Fourth, beam weight synthesis and array thinning using the *MATLAB Phased Array Toolbox* are shown. Finally, an over-the-air (OTA) calibration of a hybrid architecture is described, showing measurement and simulated performance comparisons.

II. PHASED ARRAY ARCHITECTURES

A. Passive vs. Active

An active phased array has the transmit/receive module (TRM) distributed behind each element. A passive phased array has a single TRM before the radio frequency (RF) power is split between elements.

An active phased array has improved sensitivity, lower noise figure (NF), and longer operating ranges due to increased effective isotropic radiated power (EIRP). In an active architecture, an amplifier failure will have less impact on system performance than a passive architecture, where the amplifiers are a bottleneck. Active phased arrays have increased power consumption, heat, and layout requirements, but these are trade-offs for vastly increased performance and resistance to IC failures [1].

B. Analog

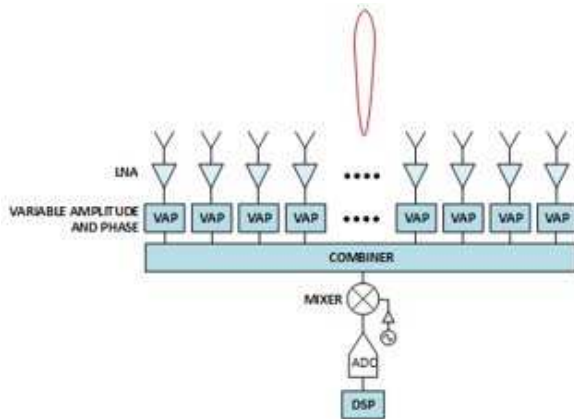


Figure 3. Active analog phased array architecture block diagram.

A block diagram of an active analog architecture is shown in Figure 3. In an analog beamforming architecture, the signal received by the aperture is stored digitally after spatial filtering. Phase and amplitude control are implemented using analog circuits. The number of analog beams is set by the number of phase control lines splitting from each element. The phase and amplitude control integrated circuits (IC) will have non-negligible quantization error depending on their bit size.

To increase analog beam count, the physical design of the system would need to change by adding additional phase control lines to each element. This requires a printed circuit board (PCB) redesign and complicates calibration and layout requirements. There is a fundamental limit on analog beam count due to the RF power being split, and also due to spatial requirements on the PCB to control each branch of components.

As a result, analog architectures and their beam-shapes are non-reconfigurable. Adaptive beamforming cannot be applied because the spatial filtering happens in the analog domain before processing can happen.

C. Digital

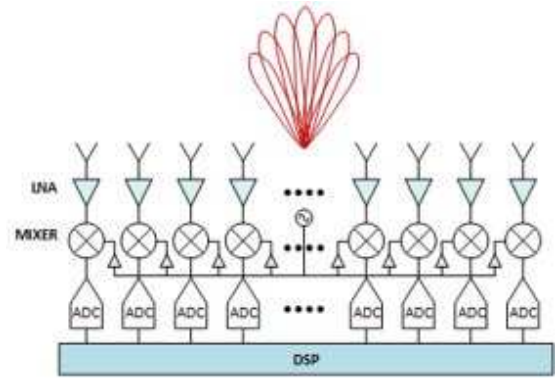


Figure 4. Digital phased array architecture block diagram.

A block diagram of a digital architecture is shown in Figure 4. In a digital architecture, each element has its own digital transceiver. The complex phasor at each element on the aperture is stored digitally, and a digital representation of the phased array aperture's state over time is available for processing. Computer precision time delay and amplitude control to each element is applied to the dataset and is limited only by the DBF processor capability and the architecture throughput limitations.

Adaptive beamforming algorithms can be implemented on the stored aperture information to achieve precise control over beam shape while accounting for unknown interferers, or to achieve other complex phased array functions.

DBF is possible because all the information at the antenna aperture is saved for each element without any spatial filtering having been applied. Because phase and amplitude control are applied digitally, the quantization error becomes negligible.

The dynamic range of a multi-receiver phased array system increases by $S_{\text{MIN}} * M^2$, where M is the number of separate receivers and S_{MIN} is the dynamic range of the individual receiver. This is called *spatial diversity*, and thus a digital phased array architecture can use transceivers with lower dynamic range while still maintaining required system dynamic range [2].

A digital phased arrays receivers sees very little to no analog gain in its channel, so its linearity is independent of the channel gain. The linearity of active-analog and hybrid architectures is limited by the IIP3 of the front-end LNA and reduces as a function of the analog channel's beamforming gain [2].

The most significant trade-off when using a digital architecture is an increase in the power consumption and heat dissipation. Digital transceivers have high power consumption relative to their analog components. Placing a transceiver behind every antenna element increases power consumption by magnitudes [3].

D. Hybrid

A hybrid architecture uses both analog and digital beamforming techniques. The hybrid architecture can have one or more analog beams and can have any number of digital channels per N analog elements, with a minimum of $N = 2$. This is called the analog-to-digital ratio (A:D). Figure 5 shows a block diagram of two hybrid architectures, both with a 4:1 A:D. The left block diagram is for a one-analog beam design, and the right block diagram is for a two-analog beam design.

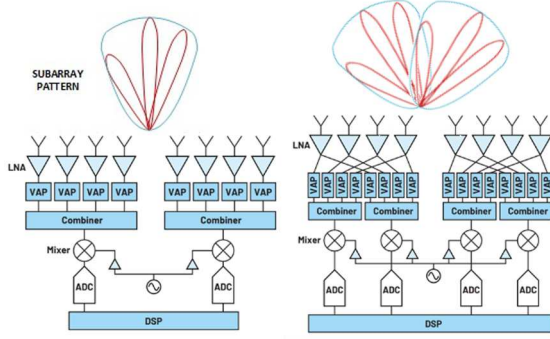


Figure 5. Hybrid phased array architecture block diagram for a 4:1 A:D ratio. The left diagram has one-analog beam, and the right diagram has two-analog beams.

Increasing the A:D ratio reduces power consumption by reducing the number of transceivers. This comes with the trade-off of increasing the distance between each subarrays digital phase center, which will result in digital quasi-grating lobes (QGL). How the A:D ratio is implemented physically determines the distance of the digital phase center between subarrays. For example, a 64:1 A:D hybrid architectures subarray can be implemented as an 8x8, 4x16, or 2x32 configuration. Each implementation will see QGL at different angular spacings depending on the scanning direction and the subarray dimensions.

Increasing the number of analog beams increases the number of required digital channels and transceivers. In a multi-analog beam hybrid architecture, the analog beams can be beamformed to overlap in the center, also called a flat-topped subarray far-field pattern. This reduces the subarray sidelobes and extends the scan space of the digital beams [2]. The digital scan space may still be limited due to QGL.

A hybrid architecture is an attractive choice when large beam counts and low power consumption are necessary. With the release of ultra-wideband (UWB) beamforming integrated circuits (BFIC), a simple analog IC schematic can be implemented on PCB to achieve hybrid or analog spatial filtering.

E. Applications

The trade-space when considering architecture selection and application requirements is massive. Some of the considerations are the coordination of analog and digital components, quantization of analog IC, calibration

implementation and limitations, hardware complexity, component costs, beam requirements, power efficiency versus beamforming accuracy, heat dissipation, interference and crosstalk between internal components, element choice and design, subarray implementation, software control, data through-put requirements, and more. High-level pros and cons of each architecture are shown in Table 1.

TABLE I. PROS & CONS OF PHASED ARRAY ARCHITECTURES

System	Pros	Cons	Applications
Analog	- Simplicity - Power	- Low Beam # - Quantization Error	- Traditional phased arrays - When simplicity and low complexity are more critical than power efficiency
Digital	- Flexibility - Precision - High Beam #	- Power - Cost	- Systems where flexibility and adaptability are crucial - When power efficiency is not a primary concern
Hybrid	- Flexibility - Power - High Beam #	- Complexity - Quantization - QGL	- Millimeter-wave phased arrays with many elements - When power efficiency and flexibility are desired

Geospatial applications require low power consumption and high beam count; thus, hybrid architectures are often used. Electronic warfare (EW) requires high beam counts, wide instantaneous bandwidth (IBW), and low latency, but does not require low power consumption; thus, digital or hybrid architectures are often used.

For multi-application use, an all-digital architecture is optimal because it has the most flexibility to cover all application spaces. However, reducing the power consumption of the digital transceivers is necessary to enable their use in low power applications such as mobile platforms. A large amount of effort is going into research and development of low power transceivers and IC development to achieve this [1]. Additionally, with increases in transceiver sampling rates, new receiver architectures can be implemented to reduce digital power consumption and layout requirements. The direct sampling receiver architecture is now practical up to X-band and above using the APOLLO chip. Direct sampling architectures use less power, have a smaller PCB footprint, and require less complex frequency planning than heterodyne and direct conversion receiver architectures [4].

III. ANTENNA SELECTION & DESIGN

A. Fractional Bandwidth

The phased array specifications for scanning range and bandwidth inform element selection. An element's fractional bandwidth (FBW) will define system BW limits. An element's embedded element pattern will define scanning loss versus scan angle.

FBW greater than 50% is considered UWB. A popular UWB element used in phased arrays is the Vivaldi element due to its simplicity of design and ease of implementation. A recent UWB antenna design is the tightly coupled array (TCA), which

makes use of high mutual coupling between elements to achieve greater bandwidths. A TCA can achieve an FBW greater than 170% [5] and can be made additively as an all-metal array [6], reducing cost, and increasing manufacturing flexibility. Additive manufacturing also enables non-conformal arrays [7]. The high mutual coupling between elements of a TCA requires greater consideration to verify active impedance as a function of scan angle to avoid scan blindness. Additionally, it is impossible for a TCA to achieve the full combination of UWB, low-height profile, high radiation efficiency, *and* high polarization purity. There is always a trade-off to be made by the TCA designer because the magnitude or phase of the reflection from the ground needs to be modified [5], [8], [9]. A TCA is generally made of dipole or Vivaldi-like elements but can be made using other elements.

Table II shows a list of antennas with their FBW and polarization capabilities. When an antenna element is capable of dual polarization, its polarized feeds can be fed in quadrature to achieve circular polarization.

TABLE II. ANTENNA FBW's

Antenna	FBW (%)	Polarization
Patch	3% to 6%	Single or Dual
Stacked Patch	10% to 25%	Single or Dual
Dipole	5% to 10%	Single
Crossed Dipole	8% to 15%	Single or Dual
Conical	>120%	Single
Crossed Conical	>120%	Single or Dual
Flared (Vivaldi)	>150%	Single
Horns	170%	Single or Dual
Spiral	180%	Circular

B. Embedded Element Pattern

The scanning range of a phased array system is estimated from the embedded element pattern of its elements. The embedded element pattern is the far-field pattern of an antenna element in its array environment. Infinite array simulations are a first step in modelling an embedded element pattern. The larger the array, the more accurate the infinite array's embedded element pattern is to determine array performance. The smaller the array, the less accurate the infinite-array solution is due to edge effects.

If the embedded element pattern has degraded loss within the scan range, significant scan loss and potentially scan blindness can occur [10].

Full-wave solvers like HFSS are used to simulate the embedded element patterns. The infinite-array technique makes use of periodic boundary conditions on a unit-cell geometry. If enough processing (RAM and CPU cores) is available, then the HFSS component-array domain decomposition method (CADDM) can be used to do a finite array simulation to account for edge-effects. CADDM can use unit-cells with arbitrary geometries, so it is used for arrays with non-uniform elements, rotated elements, and elements with non-uniform radome structures. CADDM requires more computing power than an infinite array simulation, but is more accurate, especially for smaller arrays and non-uniform arrays.

Figure 6 shows ADI OTA measurements compared to HFSS CADDM simulations for a K/Ka-band analog architecture phased array at 21GHz. The ADAR3000 TX and ADAR3001 RX BFICs are used to achieve the spatial filtering and allow up to four analog beams simultaneously. The measured embedded element pattern matches closely with the HFSS CADDM simulation, achieving a $\cos(\theta)^{2.8}$. Figure 7 shows the physical array.

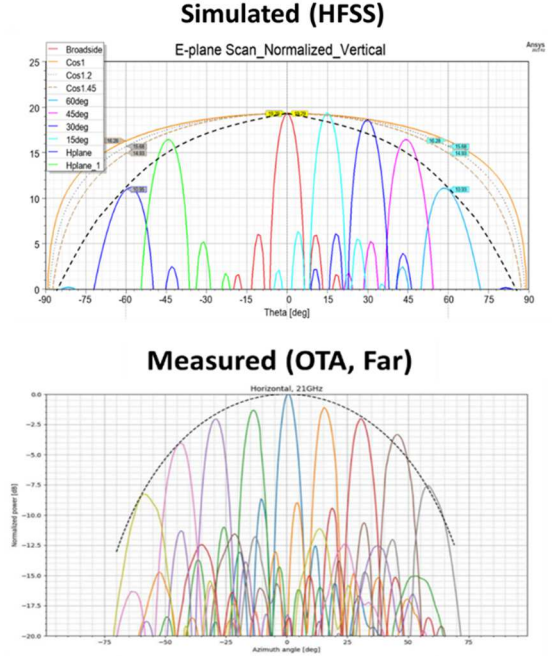


Figure 6. Simulated and measured embedded element patterns at 21GHz for an 8x8 K/Ka-band phased array. HFSS CADDM used.

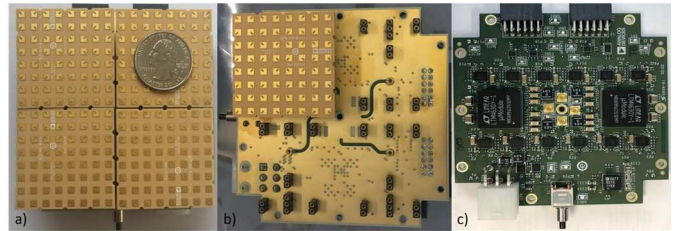


Figure 7. K/Ka-band phased array; a) Four tiles of 8x8 patch array, b) Bottom side of backplane interfacing with tiles, c) Top side of backplane with power management and SPI.

C. Mutual Coupling

Antenna designers must account for the *mutual coupling* of the array's elements. Mutual coupling is when the power from an element couples into its neighboring elements. A mutual coupling dataset is a passive S-parameter file which has a port size of N, the number of elements in the array. A 128-element array requires a 128-port SnP file to accurately model the passive coupling between elements. Alternatively, this file size can be reduced to only include M-neighbors surrounding an antenna element if coupling reduces dramatically beyond M for large arrays.

Once mutual coupling is modeled, the active impedance can be estimated through cascade analysis. The *active impedance* is the impedance seen by the TRM when looking towards the antenna element, while multiple elements are turned on simultaneously. If all elements are on in an array, then all elements will see power bleed from all other elements. This coupled power is in-phase or out-of-phase depending on the passive coupling parameters between elements. It is most optimal when most of the coupled power is out-of-phase with each other.

For a 2-port system, equations (1) and (2) show the active impedance expressions for ports 1 and 2, respectively. Variables a_1 and a_2 are phasors that represent the excitations applied to each port. The passive S-parameters, S_{11} , S_{12} , S_{21} , and S_{22} , tell us how much power from these excitations reflect and couple to each other within the system. Figure 8 shows the active impedance for all combinations of phase for a_1 and a_2 , while their magnitudes are equal. The S_{12} magnitude is increased to show the impact a higher coupling magnitude has on port 1's active impedance as a function of the excitation's phases. There is a clear maxima and minima when the coupled energy is in-phase and out-of-phase, respectively. If the passive S-parameters change, or if the excitations change in amplitude, the active impedance solution-space changes accordingly. Equation (3) is the generalized active impedance expression for any port in an N-port system.

$$\text{Active } S_1 = \frac{S_{11}a_1 + S_{12}a_2}{a_1} \quad (1)$$

$$\text{Active } S_2 = \frac{S_{22}a_2 + S_{21}a_1}{a_2} \quad (2)$$

$$\text{Active } S_{mn} = \sum_{i=1}^N \frac{S_{mi}a_i}{a_n} \quad (3)$$

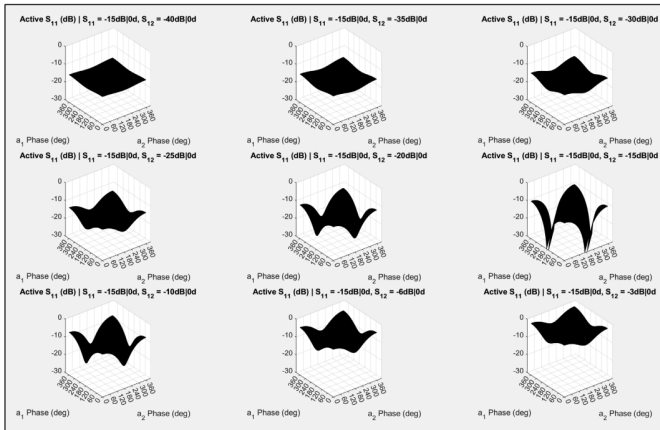


Figure 8. Active impedance of port 1 versus all phase combinations of a_1 and a_2 . Magnitude of the coupling parameter S_{12} increases from -40dB at top-left, to -3dB at bottom-right. Magnitude of $a_1 = a_2 = 10\text{dB}$. Reflection coefficient $S_{11} = S_{22} = -15\text{dB}$.

The worst-case scenario for mutual coupling is maximum in-phase contributions at the reference element from all other

elements. In this scenario, the active impedance seen by the TRM results in large return losses and massively reduced efficiency. This is called *scan blindness* due to the high active reflection coefficients. There can be a large amount of excitation states for a given mutual coupling matrix that results in high reflection coefficients.

A phased array system must not have scan blindness while operating within the expected range of scan angles (phase delta between port excitations) and amplitude tapers (magnitude of port excitations). For a multi-port system, the complexity of calculating the active impedance for all excitations increases exponentially per the number of elements. However, only certain combinations of phase and amplitude states are required to be modeled in cascade for an active impedance verification based on the applications expected scanning angles and amplitude tapers.

Applying beam weights to an antenna aperture requires the error induced by mutual coupling to be calibrated out. If mutual coupling is not accounted for, then there will be errors and the antenna aperture will not achieve the wanted beam weights.

Mutual coupling is accounted for in the embedded element pattern when using an infinite or finite array simulation [10]. Mutual coupling can be modelled in a multi-port simulation in HFSS to provide the passive S-parameter file directly. The mutual coupling file and embedded element patterns can be implemented in cascade analysis tools such as Keysight's *SystemVue* or MATLAB's *RF Toolbox* to verify active impedance performance of a system over its possible states.

IV. MODELLING SYSTEM PERFORMANCE

A cascade model of a phased array system, from bits to antenna, can estimate the system-level performance as a function of the analog cascades internal states, the different antenna arrays that be interfaced with it, and for different digital platforms on the back end.

Modelled or measured data can be used to represent each IC in the analog channel. PCB performance and coupling between transmission lines can be included as well. This type of cascade can estimate system-level performance as a function of BFIC phase shifter or TDU states, DSA states, amplifier bias settings, filter settings, and switching states. Different IC's have different states to include parametric datasets for depending on their internal complexity. A cascade requires small-signal and large-signal datasets over each IC's state. The dataset representing the cascade can become very large, but interpolation can be used to reduce the dataset size and increase cascade analysis speed.

As an example of data-set size for a single IC, the ADAR4000 BFIC has 4-analog channels, and each channel is composed of a 6-bit digital signal attenuator (DSA), 6-bit time delay unit (TDU), and 3-bit variable gain amplifier (VGA). This means there are $4 \times 64 \times 64 \times 8 = 131,072$ datasets over some defined frequency range. This is repeated for small-signal and large-signal. The number of possible states further increases

exponentially with added parametric values like temperature and supply voltage.

A multi-dimensioned parametric S-parameter file (SnPMDIF) is used to concatenate all the individual small and large-signal datasets into a single parametric dataset. An SnPMDIF can be input into SystemVue or ADS to represent linear and non-linear RF performance of a component over its internal states. Changing the state for any channel within the cascade model of the ADAR4000 will change the internal data reference to the appropriate measured dataset via the SnPMDIF dataset. If non-linearity is reached in the IC, then non-linear datasets will be used. This means we can estimate figures of merit (FOM) for any desired cascade as a function of its IC states. This also accounts for when any IC in the cascade is in its linear or non-linear region.

Important system FOM for cascades include system NF, 3rd (and 2nd) order input-referenced intercept points (ITOI), system gain, 1dB input-referenced power compression point (IP1dB), 3rd (and 2nd) order spurious-free dynamic range (SFDR3), and system dynamic range. A UWB phased array architecture will benefit from a cascade model over a wide range of frequencies, a multitude of internal system states, and over a wide IBW for applications like pulse compression and frequency modulated continuous wave radar (FMCW).

Cascades can be used to compare performance of different phased array architectures, and using measured parametric data in the cascade models will give phased array designer greater ability to ensure the system FOM will be in specification for all usable states of the selected phased array architecture.

Antenna models can be added to the phased array cascade to verify its interaction with different antenna arrays. Modelled or measured antenna performance, such as the embedded element pattern and mutual coupling parameters, can be implemented into the cascade directly.

Digital models of the transceiver, such as the APOLLO, can be implemented using MATLAB's *SerDes Toolbox*. The transceiver models can interface directly with the analog cascades, completing the full signal chain from bits to antenna. The SerDes IBIS-AMI models can be used in MATLAB, ADS and other cascade software. IBIS-AMI is a standard format for system level SerDes modelling [11].

V. BEAMWEIGHT SYNTHESIS

Beam synthesis defines the weights to be applied to each element's phasor in an antenna array to achieve a desired far-field pattern. Beam weights are applied to each element's embedded pattern in the form of amplitude and phase, and the far-field pattern is then synthesized through *superposition* of the weighted embedded element patterns [12]. This process is automated in MATLAB's *Phased Array Toolbox* which can synthesize beam weights for arbitrary antenna arrays using modelled, measured, or ideal embedded element patterns. Embedded element patterns from HFSS simulations or OTA measurements can be used to represent the element patterns of an array in the *Phased Array Toolbox*.

This tool, when combined with the RF cascade model and IBIS-AMI files, allows for estimation of the full phased array architectures performance without having to physically build the antenna and cascade. The individual ICs will need to be modelled or measured for accurate analysis.

Beam weights synthesized using modeled, measured, or ideal embedded element patterns can be implemented in the cascade model to estimate the required state to achieve those beam weights. This can also estimate calibration limits due to quantization and RMS errors.

Two examples of using the *Phased Array Toolbox* to generate beam weights are shown. For each example, a 32x32 1024-element array with 8.33mm square-lattice spacing is used. The frequency is set to 18GHz, and the 6-bit quantization error of the ADAR4000 BFIC is included. An embedded element pattern of $\cos(\theta)^{1.6}$ is assumed for all elements, ignoring edge-effects.

The first example is thinning an array using a genetic algorithm [13]. The goal is to reduce the number of elements being used to reduce array power consumption while maintaining or improving side-lobe level (SLL) performance. Figure 9 shows the thinned array after synthesis. Figure 10 shows the elevation cut, where EIRP drops by 6dBi. Table III shows the SLL and fill rate for the full, initial, and synthesized generations.

TABLE III. THINNED ARRAY - GENETIC

Generation	SLL (dBc)	Fill Rate (%)
Full	13.34	100.00
Initial	12.65	51.95
Synthesized	17.50	43.36

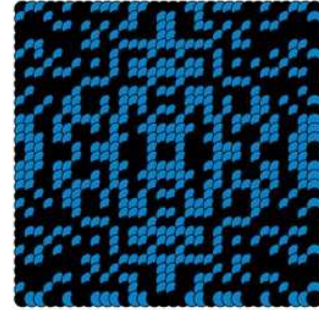


Figure 9. Thinned array after genetic algorithm is applied for 30 generations.

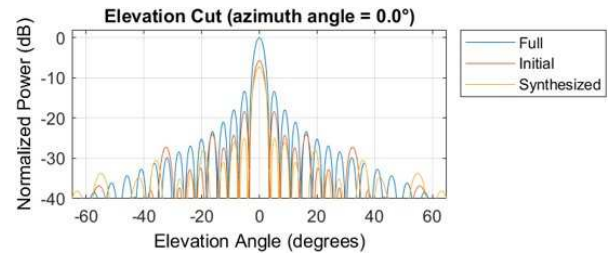


Figure 10. Elevation cut of the array for full, initial, and synthesized thinning generations. The azimuth cut looks similar.

The second example is an implementation of narrow band linearly constrained minimum variance (LCMV) beamforming [14]. The goal of LCMV is to minimize total noise output while preserving signal from a given direction. The linear constraints allow multiple simultaneous nulls.

In this LCMV example, two nulls are set at $\pm 70^\circ$ AZ simultaneously while scanning from -60° AZ to $+60^\circ$ AZ. The script is run with and without 6-bit quantization error. Figure 11 shows LCMV with 6-bit quantization error. Figure 12 shows LCMV without any quantization error. Nulls are maintained in both cases, but noise variance and SLLs perform worse for the 6-bit quantized architectures, especially at wide scan angles. The root-mean square (RMS) errors of the TDU and DSA structures in the BFICs will have a similar impact.

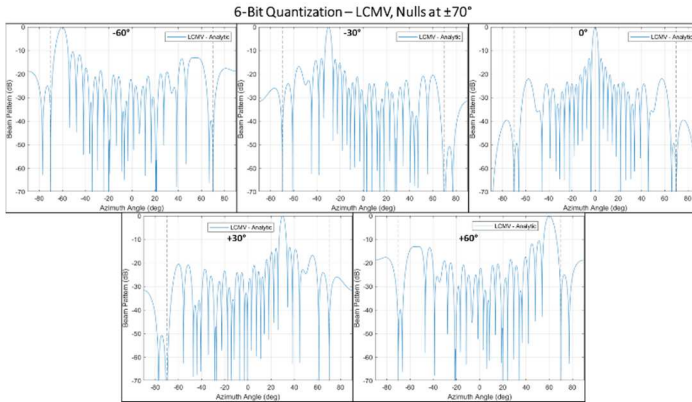


Figure 11. LCMV far-field pattern at -60° , -30° , 0° , $+30^\circ$, and $+60^\circ$ AZ scan angles, while including 6-bit quantization error.

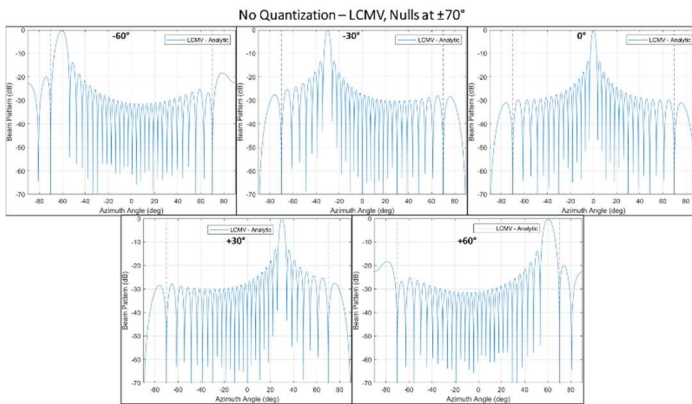


Figure 12. LCMV far-field pattern at -60° , -30° , 0° , $+30^\circ$, and $+60^\circ$ AZ scan angles, with no quantization error.

The ability to quickly synthesize beam weights given an arbitrary antenna array is useful but must ultimately be verified with OTA measurements.

VI. OVER THE AIR CALIBRATION

Calibration is a major consideration during architecture development. The levers available in the system define the limitations of the calibration.

Digital architecture levers include access to the numerically controlled oscillator (NCO) and programmable finite impulse response filter (PFIR) blocks to achieve phase and amplitude control as well as channel equalization [2]. Analog architecture levers include access to the IC phase shifters, TDUs, DSAs, VGAs, and tunable filters. A hybrid architecture has access to both sets of levers for calibration.

A basic four-step far-field OTA calibration routine is described for a hybrid architecture. The calibration is applied to a 32-element X-band ADAR1000-based 8:1 A:D hybrid architecture platform that has one analog beam. The calibration performance is verified using OTA far-field measurements [16], [17].

1. Normalize amplitude to the lowest power element: This is achieved via the VGA or DSA in the BFIC, the DSA in another cascaded component, or the bias of the TRM amplifiers.
2. Align digital phase across digital channels within a subarray: This is achieved via the NCO or PFIR and is done with respect to a natural reference element.
3. Align digital phase across subarrays: This is achieved via the NCO or PFIR and is done with respect to a natural reference element.
4. Align analog phase within a subarray: This is achieved via phase shifter or TDU in the BFIC and is done with respect to a natural reference element *per subarray*.

Some of these steps, such as 1 and 4, can be done simultaneously for calibration efficiency [18].

Figure 13 shows the far-field results post-calibration at 10GHz for a flat-taper at boresight. Figure 14 shows the far-field results for the same taper and frequency when it is scanned to $+30^\circ$. The measured results match closely with the modelled data, even for the small array size, except at wide scan angles.

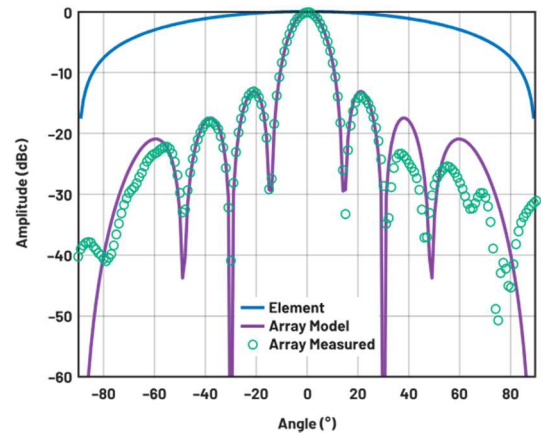


Figure 13. Measured vs simulated far-field patterns of the X-Band phased array platform at 10GHz given a flat amplitude taper and boresight scanning angle.

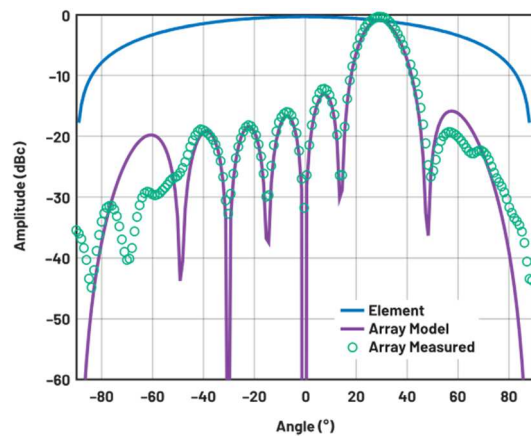


Figure 14. Measured vs simulated far-field patterns of the X-Band phased array platform at 10GHz given a flat amplitude taper and +30° scanning angle.

VII. CONCLUSION

Understanding the impact of the antenna selection and phased array architecture is important in designing a phased array system that meets desired application specifications. Knowing how to synthesize beam weights, model the architecture over its internal states to achieve beam weights, and accurately calibrate is a powerful verification step before committing to manufacturing a cascade or antenna platform. The above process can be repeated for different architectures for performance comparison before physical manufacturing. OTA testing using near-field or far-field measurements will be necessary to verify performance of the architecture, the quality of the implementation of the synthesized beam weights, and the accuracy of the calibration procedure applied. This can reduce the number of physical iterations required for phased array designs.

The benefit of going from a single application phased array to a multi-application phased array is a reduction in size, weight, cost, and power, but not necessarily performance due to wideband versus narrowband amplifier performance. If the performance of a multi-application phased array remains in-specification across its desired application spaces, then a multi-function phased array is the better design choice. Applications requiring high-performance specifications will still need to use application-specific complex architecture implementations.

REFERENCES

- [1] Conway, J. S. Herd and M. D., "The Evolution to Modern Phased Array Architectures," *Proceedings of the IEEE*, vol. 104, no. 3, pp. 519-529, March 2016.
- [2] S. H. Talisa, K. W. O'Haver, T. M. Comberiate, M. D. Sharp and O. F. Somerlock, "Benefits of Digital Phased Array Radars," *Proceedings of the IEEE*, vol. 104, no. 3, pp. 530-543, March 2016.
- [3] P. Saha, "A Quantitative Analysis of the Power Advantage of Hybrid Beamforming for Multibeam Phased Array Receivers," *Analog Devices*, 1 Dec 2021. [Online]. Available: <https://www.analog.com/en/resources/technical-articles/power-advantage-of-hybrid-beamforming.html>.
- [4] P. Delos, "An Interview with Analog Devices Discussing RF Electronics for Phased Array Applications," *Analog Devices*, 1 Jan 2019. [Online]. Available: <https://www.analog.com/en/resources/technical-articles/an-interview->

with-analog-devices-discussing-rf-electronics-for-phased-array-applications.html.

- [5] H. Tang et al., "A Broadband High-Efficiency Dipole Array Based on Frequency Selective Surface and Integrated Feeding Structure," *IEEE Open Journal of Antennas and Propagation*, vol. 2, pp. 1087-1097, 2021.
- [6] C. Pfeiffer, J. Massman and T. Steffen, "3-D Printed Metallic Dual-Polarized Vivaldi Arrays on Square and Triangular Lattices," *IEEE Transactions on Antennas and Propagation*, vol. 69, no. 12, pp. 8325-8334, Dec 2021.
- [7] C. Pfeiffer and J. Massman, "An UWB Hemispherical Vivaldi Array," *IEEE Transactions on Antennas and Propagation*, vol. 70, no. 10, pp. 9214-9224, Oct 2022.
- [8] J. P. Doane, K. Sertel and J. L. Volakis, "Bandwidth limits for lossless reciprocal PEC-backed arrays of arbitrary polarization," *IEEE Trans. Antennas Propag.*, vol. 62, no. 5, pp. 2531-2542, May 2014.
- [9] J. P. Doane, K. Sertel and J. L. Volakis, "Matching bandwidth limits for arrays backed by a conducting ground plane," *IEEE Trans. Antennas Propag.*, vol. 61, no. 5, pp. 2511-2518, May 2013.
- [10] Mathworks, "Modeling Mutual Coupling in Large Arrays Using Embedded Element Pattern," [Online]. Available: <https://www.mathworks.com/help/antenna/ug/modeling-mutual-coupling-in-large-arrays-using-embedded-element-pattern.html>.
- [11] M. Marlett, M. Tirupattur, K. Willis, and K. Keshavan, "Using IBIS-AMI in the Modeling of Advanced SerDes Equalization for Serial Link Simulation," August 2013.
- [12] MathWorks, "Verification of Far-Field Array Pattern Using Superposition with Embedded Element Patterns," [Online]. Available: <https://www.mathworks.com/help/antenna/ug/verification-of-far-field-array-pattern-using-superposition-with-embedded-element-patterns.html>.
- [13] MathWorks, "Array Pattern Synthesis Part I: Nulling, Windowing, and Thinning," [Online]. Available: <https://www.mathworks.com/help/phased/ug/array-pattern-synthesis.html>.
- [14] MathWorks, "Array Pattern Synthesis Part II: Optimization," [Online]. Available: https://www.mathworks.com/help/phased/ug/array_pattern-synthesis-part2.html.
- [15] Johnson, Don H. and D. Dudgeon, *Array Signal Processing*, Englewood Cliffs, NJ: Prentice Hall, 1993.
- [16] S. Ringwood, P. Delos, M. Jones and M. Stetzler, "Over-the-Air Pattern Measurements for a 32-Element Hybrid Beamforming Phased Array," *Analog Devices*, 9 Jun 2023. [Online]. Available: <https://www.analog.com/en/resources/journalandarticles/technical-articles/over-the-air-pattern-measurements-for-hybrid-beamforming-phased-array.html>.
- [17] S. Ringwood, P. Delos and M. Jones, "Hybrid Beamforming Transmit Calibration with SFDR Optimization," *Analog Devices*, 18 Apr 2023. [Online]. Available: <https://www.analog.com/en/resources/journalandarticles/technical-articles/hybrid-beamforming-transmit-calibration-with-sfdr-optimization.html>.
- [18] C. Fulton, M. Yeary, D. Thompson, J. Lake and A. Mitchell, "Digital Phased Arrays: Challenges and Opportunities," *Proceedings of the IEEE*, vol. 104, no. 3, pp. 487-503, March 2016.

979-8-3503-9214-2/24/\$31.00 ©2024 IEEE